AI-Driven Diabetes Management: A Retrieval-Augmented Conversational Approach*

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***Abstract*—This paper presents Diabot, a specialized conversational chatbot designed to assist individuals managing diabetes. The development of Diabot addresses a critical need for accurate, accessible, and empathetic diabetes information in a digital format. Three approaches were explored to enhance the chatbot's capabilities: prompt engineering, fine-tuning of Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG). The implementation focuses on contextual filtering, specialized LLM utilization, and an intuitive user interface. The RAG approach emerged as the most effective solution, overcoming hardware limitations and resource constraints while maintaining response accuracy. The system architecture, implementation challenges, and future improvements are discussed, including the integration of calculation capabilities for precise numerical responses. Diabot demonstrates the potential of specialized conversational AI in healthcare, particularly for chronic condition management like diabetes.**

***Index Terms*—Conversational AI, Diabetes Management, Large Language Models, Retrieval-Augmented Generation, Healthcare Chatbot, Prompt Engineering**

I. Introduction

Diabetes affects approximately 537 million adults worldwide, and projections suggest an increase to 783 million by 2045 [14]. In Morocco specifically, the prevalence rate is estimated at 10.6% among those over 20 years old, translating to approximately 2.5 million affected individuals, with over 40% unaware of their condition. This growing health concern necessitates innovative approaches to diabetes education, management, and support.

The digital transformation of healthcare has accelerated the development of AI-driven solutions, yet general-purpose chatbots often fall short in providing the specialized, accurate information required for medical domains. When individuals search for health information online, they frequently encounter misinformation or struggle to verify the reliability of sources. This can lead to unnecessary healthcare consultations driven by fear and confusion, or worse, to inappropriate self-management decisions. In this context, we developed Diabot, a specialized conversational AI system focused on diabetes care and management. Unlike general-purpose chatbots trained on broad datasets, Diabot is designed to provide precise, contextually relevant information specific to diabetes. The system aims to enhance diabetes management by offering evidence-based guidance, personalized advice, and reliable information in an accessible format.

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The rest of the paper is structured as follows: Section 2 presents the related work in healthcare chatbots and diabetes management systems. Section 3 analyzes the differences between specialized and general language models in healthcare contexts. Section 4 explores three approaches to LLM specialization: prompt engineering, fine-tuning and Retrieval-Augmented Generation (RAG). Section 5 details the system architecture of Diabot, describing its layered approach including contextual filtering, core system implementation, and user interface design. Section 6 presents the evaluation methodology and comparative analysis of the three specialization approaches. Finally, Section 7 concludes the paper with findings and discusses future research directions for specialized healthcare conversational AI systems.

II. Related work

The development of healthcare chatbots has moved quickly, shifting from simple rule-based systems to more advanced AI-driven tools. A systematic review by Laranjo et al. [1] pointed out both opportunities and risks: these systems can expand access to health services, but they also raise questions of safety and effectiveness.

Later, Montenegro et al. [2] examined in more detail the difficulties of applying such tools in real healthcare environments. They drew attention to both technical constraints and ethical concerns, insisting that domain-specific knowledge and context sensitivity are crucial. These reflections strongly shaped the way specialized healthcare chatbots should be built, particularly with respect to accuracy and user trust. Within the diabetes field, a number of solutions have appeared. Winkler et al. [4], for instance, tested a scaffolding-oriented conversational agent in online education. Their findings suggested that structured dialogues can make it easier to understand complex material.

Cappon et al. [5] also reviewed digital technologies for managing diabetes, paying close attention to continuous glucose monitoring and its interaction with decision-support mechanisms. They concluded that data integration, along with contextual awareness, is essential for effective interventions. This point had a direct impact on the way we designed our

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system.

Liu et al. [8] later examined how prompt design can guide LLMs toward better responses in healthcare contexts. Their work showed that carefully framed instructions can significantly influence output quality. Building on this perspective, Tamenaoul et al. [15] proposed a User Prompt meta-model that systematizes recurrent patterns of effective prompts for GPT-based systems. Their approach suggests that by abstracting recurring prompt patterns into a meta-framework, prompts can be crafted systematically rather than heuristically, leading to improved precision and consistency.

Finally, for Retrieval-Augmented Generation (RAG), the work of Lewis et al. [10] established the fundamental architecture that combines knowledge retrieval with generative capabilities. Follow-up work by Izacard and Grave [11] indicated that RAG improves factual accuracy while preserving conversational flow. Together, these contributions informed our decision to adopt RAG in the diabetes chatbot.

III. Specialized vs. General Language Models

Modern Large Language Models (LLMs) such as GPT-4, Claude, and LLaMA can generate fluent text and handle diverse topics with ease. Their training relies on huge and varied datasets, which explains their versatility. That same general training, however, limits their precision in highly technical areas like healthcare.

General-purpose systems frequently stumble over domain-specific vocabulary and constraints. They may produce answers that sound smooth but are misleading or even wrong. In medicine, such errors are not trivial—they can affect clinical decisions or patient self-care. Recent studies have emphasized the need for domain-specific models in healthcare to ensure accuracy and relevance in medical contexts.

Choosing between general and specialized models involves several trade-offs. General-purpose models provide broad coverage, but often at the cost of depth. Specialized models, on the other hand, tend to be more reliable within their niche but demand curated datasets, extra training, and suitable infrastructure. Outside their domain, their performance usually drops. The decision thus depends on whether the priority is breadth or depth, and on the resources available for development.

IV. Approaches to LLM Specialization

To adapt LLMs for specific medical purposes such as diabetes support, three strategies are usually considered: prompt engineering, fine-tuning, and Retrieval-Augmented Generation (RAG).

Prompt engineering. With this method, the model is left unchanged, and the effort lies in crafting inputs that steer it toward the intended response. This approach is lightweight in terms of resources and quick to apply. Yet, its success depends strongly on the quality of the prompts, and it rarely matches the depth of more resource-intensive methods.

Fine-tuning. In contrast, fine-tuning changes the model itself by retraining it on domain-specific examples. This process adjusts its internal weights so that it captures medical knowledge more accurately. The outcome is usually more precise, but the costs are high: specialized datasets, computing power, and careful monitoring of biases are all necessary.

Retrieval-Augmented Generation. RAG combines two elements: retrieval from an external knowledge source and generative processing by the model. The query first retrieves relevant passages, and then the model uses them to generate its answer. This keeps the conversation natural but grounds it in reliable data. Because the knowledge base can be updated, RAG provides a way to stay current without full retraining.

Each of these methods carries its own balance of effort, accuracy, and cost. The "right" choice depends largely on the available resources, the intended scope of the healthcare application, and the trade-off the designers are willing to make between speed and specialization.

V. System Architecture

The architecture of Diabot was designed in layers, each with a distinct purpose. At the front end lies a **contextual filter**, which acts like a gatekeeper: only questions related to diabetes are allowed through. This filtering step is vital to keep the system focused and avoid off-topic answers. Behind it sits the **core engine**, where the actual reasoning takes place. Depending on configuration, it can rely on prompt engineering, fine-tuning, or RAG to produce responses. On top of this core, a **user interface**—inspired by ChatGPT's simplicity—makes interaction straightforward, so users can ask questions and get answers without friction. We also sketched a future enhancement: a **calculation module** that would handle numeric queries with greater accuracy.

In practice, the process is sequential. First, the filter checks whether a query is truly about diabetes. If yes, it passes the query to the chosen reasoning approach (prompt engineering, fine-tuning, or RAG) to generate an appropriate response. The answer is then displayed through the interface. This modular structure lets us evaluate and swap approaches without disrupting the overall user experience.

A. Layer 1: Contextual Filtering

We needed a way to make sure the chatbot stayed focused on diabetes and did not drift into unrelated topics. For this reason, we built a filtering step using the Groq API. In practice, it works like a gatekeeper: each incoming question is checked first, and only those judged relevant to diabetes are passed on. The filter is guided by a system message that spells out its role clearly.

The instructions given to the API are simple: it only needs to decide whether a question is diabetes-related or not, and then return "true" or "false." That decision is wrapped into a JSON object, which also carries a small message back to the user. When the question is valid, the message confirms it; when not, the reply gently asks the user to try again with a diabetes-related query. This approach not only filters out irrelevant queries but also provides constructive guidance to users, helping them understand the scope of the chatbot.

During testing, this filtering mechanism achieved nearly 100%

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precision in identifying relevant queries by analyzing the context of entire conversations. The high accuracy significantly enhanced the chatbot's focus and relevance, ensuring that users receive appropriate responses only for diabetes-related inquiries. This contextual understanding is particularly important in healthcare applications, where maintaining domain focus is crucial to provide reliable information.

The Groq API was selected for this task due to its superior accuracy and effectiveness in contextual analysis. While alternative approaches, such as developing a custom sentiment analysis model, were considered, they would have required substantial training data and resources, and would likely not have achieved the same level of precision as the API-based solution.

B. Layer 2: Core System

For the core system, we explored and evaluated three distinct approaches: prompt engineering, fine-tuning, and RAG. Each method offered different advantages and challenges in developing a specialized diabetes chatbot.

1) Prompt Engineering: The prompt engineering approach involved designing an effective system prompt that defined the chatbot's role, capabilities, limitations, communication style, and workflow. This method leverages the inherent capabilities of pre-trained language models without requiring additional training or specialized resources.

We developed a comprehensive system prompt that specified the chatbot would serve as a diabetes assistant offering evidence-based, empathetic advice. The prompt detailed the chatbot's capabilities, including comprehensive knowledge about diabetes (types, symptoms, treatments), personalized guidance on diet, exercise, and medication, educational resources for understanding complex medical concepts, and problem-solving support for diabetes management challenges. To ensure safety, the text reminded users that the chatbot could not replace professional care. For emergencies, it redirected people to the right services. Privacy was also underlined: no personal health information should be stored or shared. We also set the tone. The chatbot was told to stay empathetic, speak in plain language rather than jargon, and offer a space where users could feel supported without judgment. Finally, the workflow was included: start by assessing the user's needs, then give tailored advice, encourage check-ins, and point to credible resources. Altogether, this prompt worked as a set of "rules" that guided the model's responses, even though the model itself remained unchanged. While this approach required minimal computational resources and could be implemented quickly, its effectiveness heavily depended on the quality and specificity of the prompt, as well as the capabilities of the underlying language model.

2) Fine-Tuning: In the fine-tuning attempt, we first assembled a general healthcare dataset and reformatted it into structured pairs of questions and answers. The goal was to give the model direct exposure to medical-style queries, with a focus on diabetes.

We selected the LLaMA model for fine-tuning due to its high accuracy and scalability. LLaMA offers several advantages for specialized applications, including strong performance in generating accurate, contextually relevant responses and efficient handling of a large volume of queries. To optimize resource usage during the fine-tuning process, we implemented QLoRA (Quantized Low-Rank Approximation), a technique that significantly reduces memory requirements while maintaining model performance.

The QLoRA implementation included several key components. First, we applied 4-bit quantization to the LLaMA-3 model weights, compressing them to reduce the memory footprint. Most of the pre-trained model parameters were frozen post-quantization, preventing updates during fine-tuning to conserve computational resources. We then added trainable low-rank adapter layers to capture task-specific knowledge without significantly increasing the parameter count. Finally, we implemented gradient-based fine-tuning that updated only the low-rank adapters, minimizing computational overhead.
However, the fine-tuning process encountered significant challenges that ultimately prevented its completion. We faced library conflicts that required specific versions of CUDA and PyTorch, which we resolved by creating a dedicated virtual environment with the appropriate dependencies. More critically, our local hardware proved insufficient for the demands of training, even with the optimizations provided by QLoRA. When exploring cloud-based alternatives, we found that the costs would exceed $50, presenting a financial barrier to pursuing this approach.

These constraints highlight the practical challenges of implementing fine-tuning for specialized models, particularly in resource-limited environments. Despite the potential benefits of fine-tuning for creating highly specialized language models, the hardware requirements and associated costs can make this approach inaccessible without substantial infrastructure investments.

3) Retrieval-Augmented Generation: The RAG approach combines retrieval and generation techniques to enhance response accuracy and relevance. This method addresses some of the limitations of both prompt engineering and fine-tuning by enabling the system to access and incorporate specific information without requiring extensive model modification.

Our RAG implementation consisted of three main components. First, we selected Google Gemini as the language model for generating coherent responses based on retrieved information. While various LLMs could serve this purpose, Gemini was chosen for its balance of performance and accessibility. Second, we implemented ChromaDB as our vector database for storing and retrieving vectorized data. After testing multiple vector databases including Qdrant, Chroma, and Pinecone, we selected ChromaDB for its open-source nature and effective performance in managing vectorized data. Third, we utilized HuggingFace's all-MiniLM-L6-v2 for generating text embeddings, choosing this model after considering alternatives such as OpenAI's embedding model, which was more costly and token-intensive.

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The RAG process flow begins by breaking documents into manageable chunks using RecursiveCharacterTextSplitter. These chunks are then converted into vector embeddings that capture their semantic meaning. The embeddings are stored in ChromaDB, creating a searchable repository of diabetes-related information. When a user submits a query, it is also converted into an embedding, and a similarity search is performed in the vector database to retrieve the most relevant information chunks. These retrieved chunks are then provided to the language model, which generates a response incorporating the specific information while maintaining a natural, conversational tone.

During testing, the RAG approach demonstrated excellent effectiveness in extracting accurate information from various sources, including tables and PDFs, and providing precise, contextually relevant answers. The ability to incorporate specific information from reliable sources significantly enhanced the accuracy and trustworthiness of the responses, making RAG particularly well-suited for a specialized medical domain like diabetes care.

C. Layer 3: User Interface

The user interface was designed to be intuitive and accessible, drawing inspiration from ChatGPT's successful interface design. We recognized that a clear, straightforward interface would be essential for ensuring user engagement and satisfaction, particularly given the diverse user base of a diabetes-focused chatbot.

The interface features a minimalist layout with an input field where users can enter their questions or requests. The responses are displayed in a dedicated area, presented in a readable and structured manner to enhance comprehension. We implemented a filtering mechanism that screens irrelevant queries, ensuring users receive appropriate guidance when their questions fall outside the chatbot's diabetes focus. The system also automatically generates titles for conversations, helping users organize and reference their interactions over time.

Figure 1 illustrates a use case scenario of the developed platform.

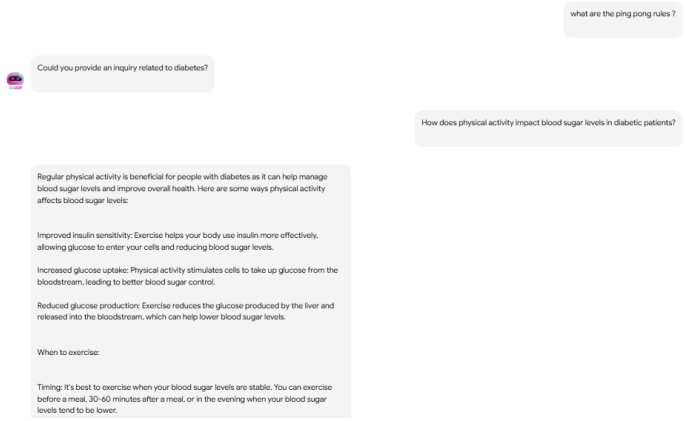


Fig. 1. Illustrative example of DIABOT platform

Diabot was developed using a stack of modern technologies selected for their robustness, performance, and suitability for AI-driven applications. For frontend development, we utilized React.js, a JavaScript library for building dynamic user interfaces. React's component-based architecture enables efficient rendering and state management, crucial for creating responsive chatbot interfaces. We enhanced React with Next.js, a framework that provides server-side rendering capabilities, improving initial load times and search engine optimization while simplifying routing and state management.

Figure 2 presents the system architecture. The backend infrastructure relies on Node.js, a server-side JavaScript runtime that offers efficient handling of asynchronous operations, making it well-suited for the real-time nature of chatbot interactions. We implemented Express.js as our web framework, leveraging its minimalist approach to provide robust HTTP utilities and middleware for handling requests and responses efficiently. For deployment, we selected Vercel, a platform optimized for frontend applications that offers continuous deployment, automatic scaling, and domain management features, ensuring high availability of the chatbot.

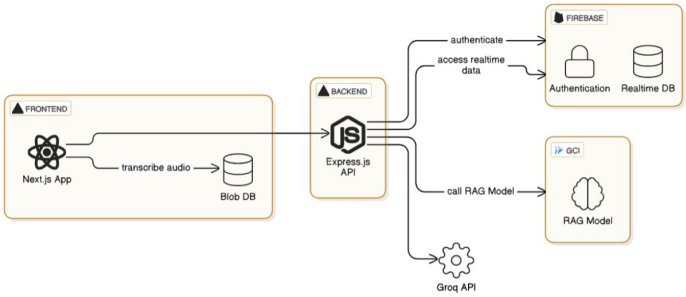


Fig. 2. System Architecture

Additionally, we incorporated Streamlit to develop a user-friendly interface for our RAG solution. Streamlit enables users to upload multiple PDFs, create embeddings from the content, and query these documents to extract valuable information without requiring advanced technical skills. This feature is particularly beneficial for extracting information from diabetes-related books and documents.

For the AI-specific components of our system, we employed several specialized technologies (Figure 3). The Groq API powers our contextual filtering mechanism, analyzing queries to ensure their relevance to diabetes. Google Gemini serves as the language model for our RAG implementation, generating coherent and contextually appropriate responses. ChromaDB provides the vector database functionality, storing and retrieving the vectorized data efficiently. Finally, HuggingFace's all-MiniLM-L6-v2 model generates text embeddings that capture the semantic meaning of both documents and queries, enabling effective similarity searching. This technology stack was carefully selected to balance performance requirements with resource constraints, enabling the development of a sophisticated chatbot system without requiring extensive infrastructure or prohibitive computational resources.

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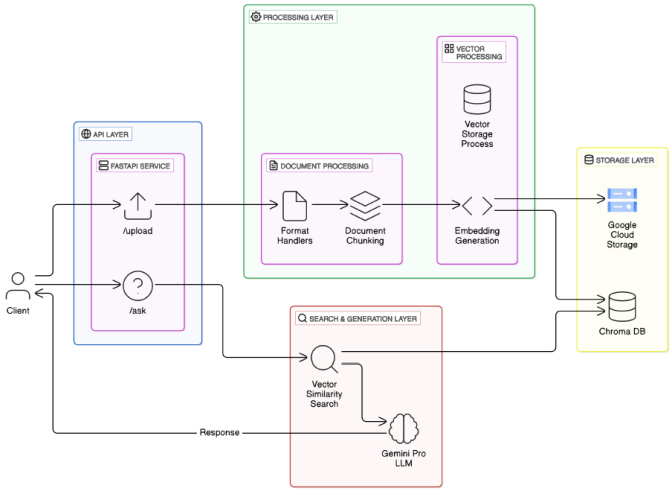


Fig. 3. RAG Architecture

VI. EVALUATION

To assess Diabot, we compared the three specialization approaches in practice. Each method was implemented and tested, which allowed us to see not only their theoretical strengths but also how they behaved under real resource constraints. This analysis considers both theoretical capabilities and real-world implementation constraints, providing insights into the trade-offs between different specialization strategies.

A. System Component Evaluation

When selecting components, we had to weigh practical aspects such as cost, performance, community support, and ease of integration. Several alternatives were tested before settling on the final stack (Table I).

TABLE I
COMPONENT SELECTION OUTCOMES

Component	Selected Component	Decision Factors
Vector Database	ChromaDB	Open-source, effective performance
Embedding Model	all-MiniLM-L6-v2[a]	Cost-effective, proven reliability
Language Model	Google Gemini	Performance-cost balance
Fine-tuning Model	LLaMA	High accuracy, scalability

[a]41M downloads on HuggingFace.

For example, in the case of the vector database, we compared Qdrant, Chroma, and Pinecone. ChromaDB turned out to be the most appropriate choice, mainly because it is open source and avoids ongoing subscription costs. A similar story applied to embeddings. Although OpenAI's model was considered, its price and token usage made it impractical. We opted instead for HuggingFace's all-MiniLM-L6-v2, which had been widely adopted by the community and offered solid performance at no extra cost. Google Gemini was chosen as the language model for the RAG implementation based on its balance of performance and accessibility. While various LLMs could serve the response generation purpose, Gemini provided the optimal combination of coherent output generation and reasonable access costs for the project scope.

B. RAG Component Analysis

The RAG approach demonstrated effectiveness across multiple document types based on testing results, establishing itself as the most viable solution for the Diabot project. The implementation process involved careful integration of retrieval and generation components to create a cohesive system capable of accessing domain-specific information while maintaining natural conversational capabilities (Table II).

Document processing utilized RecursiveCharacterTextSplitter to break down source materials into manageable chunks. This chunking strategy proved essential for maintaining semantic coherence while enabling efficient similarity searches across the vector database.

The embedding generation process leveraged HuggingFace's all-MiniLM-L6-v2 model to convert text chunks into vector representations that capture semantic meaning. These embeddings were stored in ChromaDB, creating a searchable repository of diabetes-related information that could be efficiently queried based on user input similarity.

Response generation integrated retrieved information with Google Gemini's language capabilities, producing coherent and contextually appropriate responses. The system demonstrated excellent performance in extracting accurate information from various sources, including PDFs, tables, and structured data, while maintaining the conversational flow expected in chatbot interactions.

TABLE II
RAG SYSTEM COMPONENT EVALUATION

Component	Implementation Details	
	Technology	**Performance**
Document Process	Recursive Character TextSplitter	Effective chunking
Embedding	MiniLM-L6-v2	Semantic capture
Vector Storage	ChromaDB	Efficient retrieval
Response Generation	Google Gemini	Coherent integration
Data Sources	PDFs, tables	Excellent extraction

VII. CONCLUSION

This paper presented Diabot, a specialized conversational AI system designed to provide accurate, accessible, and empathetic information about diabetes. Through the exploration of prompt engineering, fine-tuning, and Retrieval-Augmented Generation, we identified RAG as the most effective approach for our specific application, balancing accuracy and resource requirements.

The development of Diabot addresses a critical need in diabetes care, where access to reliable, personalized information can significantly impact management outcomes. By creating a specialized chatbot focused exclusively on diabetes, we aimed

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to provide a resource that offers more accurate and contextually appropriate guidance than general-purpose conversational systems. The chatbot's ability to understand diabetes-specific terminology, recognize the importance of precise information, and provide evidence-based guidance represents a meaningful advancement in AI-assisted healthcare communication.

Testing all three methods gave us insight into the trade-offs of specialized AI design. Prompt engineering was simple but shallow; fine-tuning promised depth but required heavy resources; and RAG offered a middle ground by combining retrieval with generation.

Development was not without obstacles—limited hardware and computing costs were constant concerns. Still, by carefully choosing open-source tools and efficient methods, we showed that practical and specialized AI systems can be built even in resource-constrained environments. This lesson is relevant for others aiming to build similar healthcare applications.

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